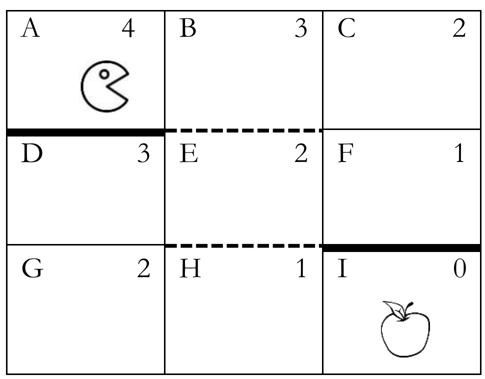
CSE 440: Artificial Intelligence
Assignment, Spring 2024
North South University
Full Marks: 40 (10 X 4)

Problem 1

Consider the very simplified version of Pacman shown below. The game world is a 3 X 3 grid where each cell is named by a letter from A to I. The Pacman agent is initially in cell A, and wants to go to cell I. Pacman has a heuristic function to estimate how close each cell is to the goal. Here are some more details:

- On the **top right corner of each cell**, the heuristic cost is given (this heuristic function was calculated using the Manhattan distance from each cell to the goal).
- From any cell, the agent can make **horizontal** (left/right) or **vertical** (up/down) moves to an adjacent cell UNLESS the border between the two cells is marked by a **thick line** (borders between A & D and F & I).
- The **cost** of an allowed move between two cells is 2 (two) EXCEPT for moves between two cells separated by a fence marked with **dashed lines** (borders between B & E and E & H) for which the move cost is 7 (seven).
-



a) Construct a search tree from this problem using the **greedy best first search** algorithm. On the tree, each node should be marked with the value of the **evaluation function** in that node, and each edge should be marked with its corresponding **weight** (the cost of that action).

b) Do the same as part (a), this time for **A* search.**

[Note: Use alphabetical order to break any ties]

Problem 2

Assume we have two predicates: **Parent(p, q)** which means "p is the parent of q" and **Female(p)** which means "p is a female". We also have constants **Fatima** and **Anwar**,

Express each of the following sentences in first-order logic. (You may use the abbreviation ∃1 to mean "there exists exactly one.")

i. Fatima has a daughter (possibly more than one, and possibly sons as well).
ii. Fatima has exactly one daughter (but may have sons as well).
iii. Fatima has exactly one child, a daughter.
iv. Fatima and Anwar have exactly one child together.
v. Fatima has at least one child with Anwar, and no children with anyone else.

Problem 3

This exercise uses the function **MapColor** and predicates **In(x, y)**, **Borders (x, y)**, and **Country(x)**, whose arguments are geographical regions, along with constant symbols for various regions. In(x, y) means "x is in y" (e.g., In(Dhaka, Bangladesh)). Borders(x, y) means that "countries x and y have borders" (e.g., Borders(Bangladesh, India). Country(x) means that "x is a country".

Express each of the following sentences in first-order logic.

i. Paris and Marseilles are both in France.
ii. There is a country that borders both Iraq and Pakistan.
iii. All countries that border Ecuador are in South America.
iv. No region in South America borders any region in Europe.
v. No two adjacent countries have the same map color.

Problem 4

Describe one problem faced by the residents of Dhaka city that you think can be solved using Artificial Intelligence. Your solution must be specific, i.e., describe what data will be collected and how, what type of algorithm will be used, how your solution will be tested and deployed, what challenges you may face, etc.